

SALES & OPERATIONS AUDIT

Northwind Sales & Operations Audit

Uncovering hidden patterns in revenue, shipping, and customer behavior to transform data into strategic decisions.



The Challenge

"Our sales grew in 2014, but our management team is flying blind. We don't know who our most loyal customers are, which regions are struggling, or if our shipping partners are delivering on time."

Missing data in shipping logs raised concerns about warehouse operations. This audit was commissioned to bring clarity.

Data Structure & Scope

Thousands of transaction records across **20+ fields**, spanning Order IDs 1001–1432+.



Order Details

Order ID, Order Date,
Shipped Date



Customer Info

Name, Address, City,
State, Country



Shipping

Shipper Name, Ship
Address, Ship City



Products

Name, Category, Unit
Price, Quantity, Sales

Key Insights at a Glance



Regions & Sales Team

- **4 Regions:** West, East, North, South (US-based)
- **Payments:** Cash, Check, Credit Card
- **Shippers:** Company A, B, C

Top Categories

Beverages, Dried Fruit & Nuts, Candy & Baked Goods, Soups, Sauces, Condiments, Pasta, Dairy, and more.

The Shipping Bottleneck Problem

"Are we getting slower at shipping? Which shipping company is the most reliable?"



Convert Dates

Order & Shipped dates to date-time objects



Calculate Lead Time

Shipped Date minus Order Date



Group & Compare

Average delay by shipper



Result: All three shipping companies have a consistent average lead time of **2.0 days**. No bottleneck from the carriers' side.

A	
name	A
Company A	
Company E	
Company C	

Salesperson Performance Mystery

"Is the West region doing better because of the market, or because of Mariya?"

	A	B	C
1	Salesperson	Region	Sales
2	Anne Larsen	South	72163.25
3	Andrew Cencini	East	54957.4
4	Nancy Freehafer	North	51325.75094
5	Laura Giussani	East	39943.51
6	Mariya Sergienko	West	39546.75094
7	Michael Neipper	North	38540
8	Robert Zare	West	32530.6
9	Jan Kotas	West	17041.5

Key Findings

- **Top Performer:** Anne Larsen (South) — over \$72,000 in sales
- **West Region:** Mariya Sergienko significantly outperforms colleague Jan Kotas

Revenue was aggregated by both Salesperson and Region, then ranked by total revenue.

Product Portfolio Audit

"What is our Hero Category, and what should we stop selling?"

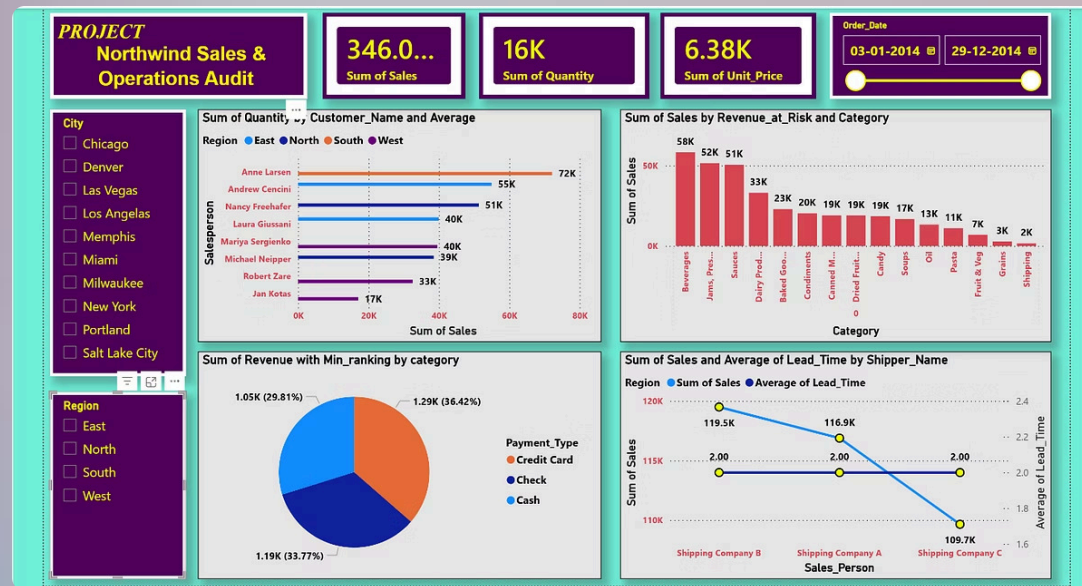
Hero Categories

Beverages and **Jams/Preserves** lead in total revenue.

Under Review

Grains is the lowest revenue category — consider discontinuation.

	A	B	C	D
1	Category	Sales	Quantity	Unit_Price
2	Beverages	58258.91094	2550	23.46533333
3	Jams, Preserves	51541	917	51.52631579
4	Sauces	50520	1263	40
5	Dairy Products	33129.6	952	34.8
6	Baked Goods & Mixe	22955.40094	2016	11.40606061
7	Condiments	20278	1207	16.6
8	Canned Meat	19117.6	1039	18.4
9	Dried Fruit & Nuts	19023.5	1075	21.71428571
10	Candy	18528.25	1385	13.09259259
11	Soups	16829.6	1744	9.65
12	Oil	13322.4	624	21.35
13	Pasta	11056.5	567	19.5
14	Fruit & Veg	6942	178	39
15	Grains	2884	412	7
16	Shipping	1662	66	23.33333333



TASK 4

Payment Preference Analysis

"Do Credit Card customers spend more than those paying by Check?"

\$1,285

Credit Card AOV

Highest average order value

\$1,146

Check AOV

Second highest basket size

\$945

Cash AOV

Lowest average spend

OBSERVED PATTERNS

Data Quality & Product Trends

High-Frequency Products

Coffee, Green Tea,
Chocolate Biscuits Mix,
Curry Sauce, Clam
Chowder, Crab Meat

Price Range

From \$2.99 (Green Tea)
to \$81 (Marmalade) —
wide spread across
categories

Order Volumes

Vary widely from ~10
units to 100+ units per
order

Duplicate Entries

Some orders (e.g., ID
1003 — Dried Pears)
appear multiple times,
suggesting repeated
or split shipments

Summary & Recommendations

O1

Shipping is Healthy

All carriers average 2.0-day lead time — no bottleneck detected.

O2

Invest in Top Talent

Anne Larsen and Mariya Sergienko are clear revenue drivers in their regions.

O3

Double Down on Heroes

Beverages & Jams lead revenue. Review Grains for discontinuation.

O4

Encourage Credit Cards

Credit Card AOV (\$1,285) is highest — incentivize card payments.

O5

Fix Data Quality

Resolve duplicate entries and missing shipping logs to ensure reliable reporting.